IR

1

1.1

The probability of a word appearing in a relevant document can be calculated from its probability to appear in each document weighted by the query's probability to have been generated by that document.

$$P(w \mid R) \propto \sum_{d \in D} P(w \mid d) \cdot \prod_i P(q_i \mid d)$$

1.2

We average the probability that each word appears in each document.

$$P(\text{cat} \mid R) = \frac{1}{3}(\frac{1}{5} + \frac{1}{8} + \frac{6}{6}) = \frac{53}{120}$$

For any word $\neq$ cat the last document contributes 0.

$$P(\text{dog} \mid R) = \frac{1}{3}(\frac{1}{5} + \frac{1}{8}) = \frac{13}{120}$$

For any $w \in \{\text{cow}, \text{pig}, \text{horse}\}$ the first document gives $\frac{1}{5}$ and the rest 0, thus

$$P(w \mid R) = \frac{1}{3} \cdot \frac{1}{5} = \frac{8}{120}$$

For any other w the first and last documents contribute 0. If the word is not "the", then d_2 contributes $\frac{1}{8}$ and

$$P(w \mid R) = \frac{1}{3} \cdot \frac{1}{8} = \frac{5}{120}$$

Finally

$$P(\text{the} \mid R) = \frac{1}{3} \cdot \frac{2}{8} = \frac{10}{120}$$

1.3

$$P(\text{cat} \mid R) = 0.7 \cdot \frac{53}{120} + 0.3 \cdot \frac{1}{2} = \frac{551}{1200}$$
$$P(\text{dog} \mid R) = 0.7 \cdot \frac{13}{120} + 0.3 \cdot \frac{1}{2} = \frac{271}{1200}$$

The rest of the words get 0 for the query part. For $w \in \{\text{cow}, \text{pig}, \text{horse}\}$

$$P(w \mid R) = 0.7 \cdot \frac{8}{120} = \frac{56}{1200}$$

For other $w \neq$ the

$$P(w \mid R) = 0.7 \cdot \frac{5}{120} = \frac{35}{1200}$$

And

$$P(\text{the}) = 0.7 \cdot \frac{10}{120} = \frac{70}{1200}$$

1.4

We would (possibly) smooth it and add it as a document to the $\sum_{d \in D}$ calculation, but with weights so it still sums to 1, i.e.

$$P(w \mid R) = (1-\lambda) \sum_{d \in D} P(w \mid d) \cdot \prod_i P(q_i \mid d) + \lambda P(w \mid q)$$

We get RM3 with a smoothed version of the query.

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Approach 1

Simply redefine the collection to consist of passages and use our favourite smoothing to define the language model. The main problem in this approach is the sparsity of the vocabulary given very short passages, which would make most words get the smallest value and would ruin the estimation.

Approach 2

Smooth with both the document and the collection, i.e.

$$P(w \mid \theta_g) = \sum_{X \in \{g, D, C\}} \lambda_X \frac{\text{tf}_{w,X}}{l(X)}$$

Where g, D, C are, respectively, the passage, document, and collection, $l(X)$ denotes the length of X, tf is the term frequency and the λs are chosen s.t. $\sum \lambda_X = 1$.
This helps against the vocabulary mismatch since even though a lot of words

would get near 0 probabilities w.r.t. passages, they would still have their weights (and meanings) borrowed from the document and collection.
Note that this is the same as smoothing with the document then smoothing the result with the collection.

Approach 3

Do the same as the previous approach (maybe replace JM with Dirichlet for some extra novlty) but have short passages contribute less to further reduce the effects of vocabulary mismatch:

$$P(w \mid \theta_g) = \lambda_g \frac{\mathrm{tf}_{w,g}}{l(g)^{1.1}} + \sum_{X \in \{D,C\}} \lambda_X \frac{\mathrm{tf}_{w,X}}{l(X)}$$

or more generally

$$P(w \mid \theta_g) = \lambda_g \frac{\mathrm{tf}_{w,g}}{l(g)^{\alpha}} + \sum_{X \in \{D,C\}} \lambda_X \frac{\mathrm{tf}_{w,X}}{l(X)}$$

so we can tune α.